$$qP_q^{(n,n)}(t) - t\frac{d}{dt}P_q^{(n,n)}(t) = -\frac{n+q}{2}P_{q-2}^{(n+1,n+1)}(t), \tag{2.10}$$

$$qtP_q^{(n,n)}(t) + (1-t^2)\frac{d}{dt}P_q^{(n,n)}(t) = (n+q)P_{q-1}^{(n,n)}(t). \tag{3.38}$$

$$\begin{aligned}\partial_x F_{3,n}^{(q)} = {} & r^{q-2}\left(qP_q^{(n,n)}\left(\frac{x_3}{r}\right) - \frac{x_3}{r}P_q^{(n,n)\prime}\left(\frac{x_3}{r}\right)\right)\underline{x} \\ & + r^{q-1}\left(q\frac{x_3}{r}P_q^{(n,n)}\left(\frac{x_3}{r}\right) + (1-(\frac{x_3}{r})^2)P_q^{(n,n)\prime}\left(\frac{x_3}{r}\right)\right)e_3. \end{aligned} \tag{3.37}$$

(2.10) (3.38)

$$= r^{q-2}\left[q P_q^{(n,n)}\left(\frac{x_3}{r}\right) - \frac{n+q}{2}P_{q-2}^{(n+1,n+1)}(t)\right] x$$

$$- q P_q^{(n,n)}(t)\Big] x$$

$$+ r^{q-1}\left(q\frac{x^3}{r}P_q^{(n,n)}\left(\frac{x_3}{r}\right) + (n+q)P_{q-1}^{(n,n)}(t)\right.$$

$$\left. - q\frac{x_3}{r}P_q^{(n,n)}(t)\right)$$

$$= -r^{q-2}\frac{(n+q)}{2}P_{q-2}^{(n+1,n+1)}\left(\frac{x_3}{r}\right)x$$

$$+ r^{q-1}(n+q)P_{q-1}^{(n,n)}\left(\frac{x_3}{r}\right)$$

$$= (n+q)\left[r^{q-1} P_{q-1}^{(n,n)}\left(\frac{x_3}{r}\right) - \frac{1}{2} r^{q-2} P_{q-2}^{(n+1,n+1)}\left(\frac{x_3}{r}\right) x \right]$$

$$= (n+q)\left[F_{3,n}^{(q-1)}(x) - \frac{1}{2} F_{3,n+1}^{(q-2)}(x)\, x \right]$$